



Bournemouth University
National Centre for Computer Animation

Measuring What Compositing Does: Developing an Evaluation Framework for Product- Preserving Advertisement Generation

Huang Langhui

Student number: s5088157

Supervised by Hammadi Nait-Charif

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github.com/HuangLanghui/Master-project-S5088157

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Abstract

Master of Science

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Generative image models can now produce advertising imagery cheaply, but they synthesise the whole canvas, including the product. Advertising has a requirement that ordinary image synthesis does not: the product must appear as the same object across every format in a campaign, with its label, logotype and proportions unchanged. This dissertation develops a pipeline that separates the two concerns, generating the background while holding the product to a reference recorded before composition, and then asks how a system of that shape can be evaluated.

Evaluation proved to be the more demanding half of the work. Because the pipeline is designed not to alter product pixels, conventional fidelity measures sit near their ceiling whatever the system does, so on their own they say very little. Measuring the other side of the problem, how convincingly a preserved product sits in a generated scene, needed measures I could not find in usable form, so I wrote two. Both turned out, partway through the study, to be responding to something other than what I had named them for. I diagnosed and corrected both, and repeated the affected experiments. A third problem, more general, lies in the standard fidelity measures themselves: when the reference against which fidelity is computed comes from an earlier stage of the same pipeline, a failure in that stage leaves the scores unchanged and undetectable.

I applied the corrected framework to 5,184 rendered advertisements covering 24 products, three random seeds, two background backends, seven relighting treatments and a six-arm ablation. Within this dataset, diffusion-based relighting altered product appearance to a degree that appears difficult to reconcile with advertising use, losing a substantial proportion of fine detail in every paired comparison. Among the compositing components, the contact shadow contributed most. The parametric relighting method the study was originally designed around produced a consistent but small improvement in scene integration, at a small and very reliable cost in fidelity. Everything here is bounded by the size and composition of the dataset and by the absence of human evaluation; Section 6.2 says by how much.

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Chapter I

Introduction

1.1 Background and motivation

A product advertising campaign is rarely a single image. The same item appears in a square social post, a vertical story, a wide web banner, and often in print, and in each of these it must be recognisably and precisely the same product. This is partly a branding requirement and partly a legal one: in most jurisdictions the depiction of a product in advertising is treated as a claim about that product, so an image showing a label that does not exist, or proportions that are not the manufacturer's, is not simply a poor image but a potentially actionable one.

Producing this imagery is expensive. A studio shoot for one product across several formats represents a meaningful cost for a small brand, and the cost recurs whenever packaging changes or a new format is required. Generative image models are an obvious attraction here, and systems built on latent diffusion (Podell *et al.*, 2024; Rombach *et al.*, 2022) can produce plausible advertising-style imagery from a short description at negligible marginal cost.

The commercial response to that economics is already visible. Product-scene generation is now an advertised feature of mainstream marketing tooling, and a market of dedicated services will place an uploaded pack shot into a synthesised setting in seconds. Architecturally these services do the thing this dissertation studies, preserving a product while generating its surroundings, but what they publish is the capability, not the evaluation: how a buyer should tell a usable render from a defective one is left to inspection by eye, at exactly the volume of imagery where inspection by eye stops scaling. That gap between deployed capability and published measurement is the practical motivation for treating evaluation, rather than generation, as the research problem.

The difficulty is that these models synthesise the entire canvas. A model conditioned on “a bottle of perfume on a marble surface” generates the bottle as well as the marble, and the bottle it generates is not the client's bottle. It holds no internal representation of a particular physical object, so prompting for consistency does not solve the problem; it produces a different plausible bottle each time. Where the subject is generic this is unproblematic. For product advertising it removes the property the image most needs to have.

1.2 Research questions

The architectural response is straightforward, and it is where this work starts: generate the background, but not the product. Extract the product from an existing photograph, place it into a synthesised scene, and constrain the pipeline so that the product's pixels never pass through a generative model. What I wanted to know was what such a system costs, and how you would tell whether it was working.

Three questions organise the dissertation.

RQ1. Can a pipeline that generates backgrounds while preserving product pixels exactly produce usable multi-format advertising, and what does the preservation constraint cost in terms of how well the product sits in its new surroundings?

RQ2. If that constraint is relaxed deliberately, by relighting the product to match its generated scene, how does product fidelity trade against scene integration, and is there an operating point where the trade is favourable?

RQ3. What is required to measure either of the above? Existing fidelity measures are near their ceiling by construction in a system of this design, and it is not initially clear what should play the role of a scene-integration measure.

I expected RQ3 to be a short preliminary. It became most of the work, for reasons set out in Section 5.5.

1.3 Approach and scope

The system built for this study is a twelve-stage pipeline that extracts a product, derives a brand palette from it, plans a layout for each target format, generates or constructs a background, composites the product with a shadow and edge lighting, and evaluates the result. The preservation property is stated as a contract and enforced by an automated test, so a change that violates it breaks the test suite instead of quietly invalidating every number in Section 5.

The empirical work covers 24 products stratified by how difficult they are to separate from their original backgrounds, three random seeds, two background generation methods, seven relighting treatments and a six-arm component ablation. All comparisons are paired within product, seed, backend and format.

The scope is deliberately narrow in several respects, and that narrowness bounds every conclusion drawn later. The dataset is small and comes from a single stock-photography source. No human evaluation was carried out, so

all judgements about how images look are mediated by automated measures, two of which I wrote for this study and validated only against controlled manipulations, never against people. The generative relighting arm was examined at one strength setting across the full dataset. Section 6.2 revisits all of this.

1.4 Structure

Section 2 situates the work in relation to image harmonisation, matting and diffusion-based generation. Section 3 describes the system. Section 4 sets out the dataset, the measures and the analysis approach. Section 5 presents results, opening with a qualitative overview of the rendered output and continuing with the paired analysis, including an account of the measurement problems encountered and how they were resolved. Section 6 discusses what the results mean, why they came out as they did, and what they do not support. Section 7 concludes.

Chapter II

Related work

2.1 Compositing and harmonisation

Placing a foreground object into a background it was not photographed in is a long-standing problem in image editing. The dominant modern approach is harmonisation, which adjusts the foreground's appearance so that its colour and illumination statistics match the new background. DoveNet (Cong *et al.*, 2020) is representative, introducing both a domain-verification training approach and the iHarmony4 benchmark, which pairs synthesised composites with the real images they were derived from.

The approach has a longer history than its deep-learning phase. Multi-scale image harmonization (Sunkavalli *et al.*, 2010) matched foreground statistics to the background across a pyramid decomposition, transferring contrast, texture and noise at several scales at once, and its limits are what motivated learned replacements: hand-designed statistics cannot know which appearance differences matter semantically. learned replacements. Deep Image Harmonization (Tsai *et al.*, 2017) trained an end-to-end network on synthesised composites and established the data recipe the field still uses: take a real photograph, perturb the appearance of a segmented foreground, and train the network to undo the perturbation. DoveNet inherited that recipe, and iHarmony4 standardised it into the benchmark most subsequent work reports against. Table 2.1 summarises where these methods sit relative to the present work, and the column that matters most is the second: every harmonisation method in the table treats the foreground as modifiable, which is precisely what product advertising forbids.

TABLE 2.1: *Approaches to making a composite look real, and where this work sits.*

Approach	Foreground	Evaluation reference	Primary measure
Multi-scale harmonisation (Sunkavalli <i>et al.</i> , 2010)	modified (statistics matched)	none; visual comparison	qualitative realism
Deep harmonisation (Tsai <i>et al.</i> , 2017; Cong <i>et al.</i> , 2020)	modified (learned)	paired real image (iHarmony4)	reconstruction error
Distribution measures (Heusel <i>et al.</i> , 2017)	not applicable	corpus of real images	feature-distribution distance
This work	preserved exactly	pipeline-recorded reference	paired fidelity and scene integration

Harmonisation and the present work address the same visual problem from opposite directions. Harmonisation modifies the foreground so that it fits; product advertising requires the foreground to remain exactly what it was. Quantifying that tension is essentially what this dissertation attempts. The relighting stage described in Section 3.2 is a controlled and deliberately limited move in the harmonisation direction, and I measure what the move costs instead of assuming it is affordable.

2.2 Matting

Separating the product from its original background is the pipeline's first substantive commitment, and the quality of that separation limits everything downstream. Two approaches are used. U²-Net (Qin *et al.*, 2020) is a salient-object detection architecture whose nested U-structure builds on the encoder–decoder design of U-Net (Ronneberger *et al.*, 2015); it underlies the widely used rembg implementation (Gatis, 2020) employed here. GrabCut (Rother *et al.*, 2004) is a classical interactive graph-cut method, used as an offline fallback requiring no learned model.

Learned matting did not begin with saliency. Deep Image Matting (Xu *et al.*, 2017) framed the task as estimating a full alpha matte from the image together with a user-supplied trimap, a rough three-way labelling of foreground, background and uncertain boundary, and most of the accuracy gains in the matting literature since have kept that interactive prior in some form. The methods used here sit on the automatic side of that line. U²-Net needs no trimap, which is what makes an unattended pipeline possible, and the price of giving up the prior is paid exactly where a trimap would have helped: the styled-scene stratum, where Section 5.8 finds the lowest extraction quality in the dataset.

The comparison between them mattered for a reason beyond output quality, developed in Section 5.5: their failure modes differ in whether the evaluation can detect them at all.

2.3 Diffusion models for background generation

Latent diffusion (Rombach *et al.*, 2022) made high-resolution text-to-image synthesis computationally practical by operating in the latent space of a pretrained autoencoder, and SDXL (Podell *et al.*, 2024) scaled the approach substantially. Both are used here for background generation, through the released SDXL Base 1.0 weights (Stability AI, 2023).

Sampling speed is a practical concern when generating thousands of images for an experimental grid, and distillation offers a large improvement: SDXL-

Turbo, produced by adversarial diffusion distillation (Sauer *et al.*, 2023), reaches usable quality in roughly four denoising steps rather than several dozen. The choice matters for this study's economics: the difference between four denoising steps and several dozen is the difference between an experimental grid that finishes in an afternoon and one that runs for days, and the caching of Section 3.5 only helps with repetition, not with first contact. It achieves this partly by removing the need for classifier-free guidance (Ho & Salimans, 2021), the technique allowing a conditional diffusion model to be steered by combining conditional and unconditional predictions. Section 5.6 describes a consequence of that removal which is relevant to this application and does not appear to be widely discussed.

2.4 Perceptual quality measures

SSIM (Wang *et al.*, 2004) remains the standard structural fidelity measure but is known to correlate imperfectly with human judgement. LPIPS (R. Zhang *et al.*, 2018) addresses this by learning a perceptual distance from a large corpus of human two-alternative comparisons, and DISTS (Ding *et al.*, 2022) extends the idea with explicit tolerance to texture resampling.

A separate family of measures evaluates generation at the level of distributions rather than of images. The Fréchet Inception Distance (Heusel *et al.*, 2017) compares the statistics of deep features extracted from a set of generated images against those of a set of real ones, and it has become the default report for generative image models. It is unsuitable here for two structural reasons. It requires samples in the hundreds before its covariance estimates stabilise, where this evaluation must score one render against one reference; and it is deliberately insensitive to the identity of any individual image, where product preservation is a property of exactly one object in one render. Both properties are virtues for judging a generator and disqualifying for judging a composite, which is why every measure in Chapter IV is paired and image-level.

I use LPIPS specifically because its validity derives from human data. Since I ran no human evaluation of my own, every claim about perceptibility here rests on that borrowed validity, and it is worth being precise about what the borrowing provides. LPIPS was calibrated on judgements of distortion similarity, not on judgements of whether a composite advertisement looks real. Applying it to the second question is an assumption: reasonable, but untested in this context. Section 6.2 returns to it.

2.5 Product datasets

Amazon Berkeley Objects (Collins *et al.*, 2022) provides catalogue imagery with metadata at a scale far beyond what was collected here. I considered it

and did not use it. The variable I needed to stratify on, how hard a product is to separate from its original background, is not annotated in existing corpora and is not straightforwardly derivable from them. The absence is structural rather than an oversight: extraction difficulty is a property of the relationship between a product and the background it happens to have been photographed against, not of the product itself, so it cannot be inherited from object-level metadata and has to be assigned by looking at the photographs. A smaller purpose-collected set carrying that annotation seemed more useful to me than a larger one without it, though Section 6.2 notes what the choice cost.

2.6 Relighting

Diffusion-based relighting has advanced quickly, and methods such as IC-Light (L. Zhang *et al.*, 2025) can impose specified illumination on a subject with convincing results. The finding reported in Section 5.4 is not that such methods produce poor images. It is that when applied to a product whose identity must be preserved, the changes introduced by the particular approach tested extend beyond illumination in ways that appear difficult to reconcile with the requirements of advertising. Relighting occupies the middle of the axis that Table 2.1 lays out: it modifies the foreground, as harmonisation does, but restricts the modification to illumination, and the question this dissertation asks of it is whether that restriction can be enforced in practice, that is, whether a method asked to change only the light in fact changes only the light. Section 5.4 answers that question for the image-to-image approach, and the answer is no.

Chapter III

System design

3.1 Pipeline structure

The system is organised as twelve stages, each with a single responsibility, summarised in Table 3.1 and illustrated in Figure 3.1.

TABLE 3.1: *Pipeline stages and their roles.*

Stage	Role
1 Understanding	product profile: category, audience, scene keywords
2 Extraction	RGBA cut-out and binary mask
3 Brand	palette, accent and neutral colours derived from product pixels
3b Copywriting	headline and tagline
4 Layout	per-format product box and copy anchor
5 Background	background at the target format resolution
6 Preservation	reference pixels recorded before composition
6b Lighting	one key-light direction shared by Stages 5 and 7
6c Relighting	optional product relighting, the study's independent variable
7 Composition	finished advertisement
8 Campaign	all three formats
9 Evaluation	metric report

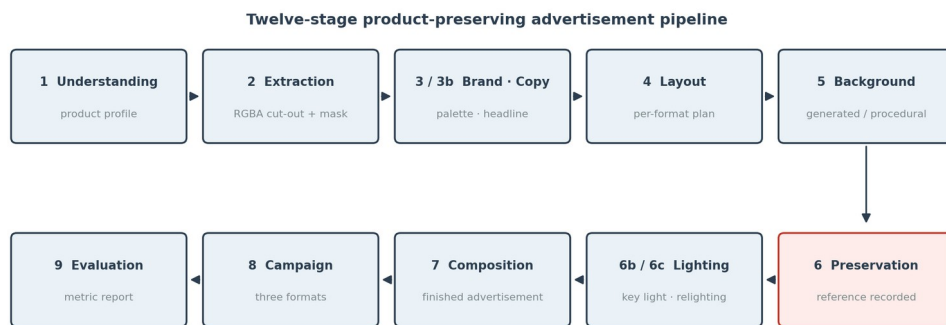


FIGURE 3.1: *The twelve-stage pipeline. Stage 6 (highlighted) records the preservation reference before composition begins, which is what makes the fidelity measurements in Section 5 interpretable.*

3.2 The preservation contract

Between extraction and the finished advertisement, the only operations applied to product pixels are a single LANCZOS resample to the planned

size and alpha blending at the silhouette boundary. Stage 6 records the post-resample pixels as a reference before composition begins, so that Stage 9 compares the rendered product against what it should look like at that size, not against the original photograph. It therefore measures composition damage and does not penalise the intended resample.

Stage 6c relights the product after that reference is captured. The ordering is what makes the study feasible: the reference stays the unmodified product, so the fidelity drop Stage 9 reports is by construction the cost of the relighting, and I needed no separate instrumentation. The relighting module (`relight.py`) offers four modes: none; graded, a 2.5-D Lambertian shading derived from a pseudo-normal field obtained by distance transform of the silhouette; resample, a control described in Section 5.4; and generative, diffusion image-to-image applied to the product region.

The preservation property is expressed as an automated test that composes an advertisement and asserts byte equality between the fully opaque product pixels and the recorded reference. The test earns its place. Several components added during the project draw close to the product boundary, and without it I could easily have shipped a change that invalidated every fidelity number in this dissertation while producing no visible symptom at all. The test asserts byte equality rather than similarity above a threshold, and the choice is deliberate. A tolerance invites drift: every change that stays inside it is invisible, and tolerances compose, so three individually tolerable changes can sum to an intolerable one that no single commit ever triggered. Byte equality composes the other way, since the only way to pass is to touch nothing, which turns the preservation property from a quantity to be monitored into an invariant to be maintained.

3.3 Lighting and composition

A single key light is planned per campaign and consumed by both the background prompt and the compositing, so that the backdrop's brightness falloff, the contact shadow and the rim light agree about where the light comes from. The contact shadow is constructed from the product's own silhouette, inverted, flattened according to the light's elevation and sheared along its direction. The more common approach, an ellipse beneath the bounding box, bears no relation to the object's shape, and it is one of the things that makes a composite read as a paste-up.

Copy colour is selected by WCAG contrast against the region the text will actually occupy, and a blurred scrim is placed behind the text when neither light nor dark ink reaches adequate contrast. The scrim is drawn before the product is composited, so it cannot affect preserved pixels even if a future layout change brings text and product closer together.

3.4 Copy and layout

Stage 3b resolves the headline and tagline through a provider chain in which an explicit line always takes precedence: copy supplied in configuration is used unchanged; otherwise a vision-language model proposes variants from the product image and the earlier profiles; otherwise deterministic templates keyed to the product profile are used. The templates are separated into generic and category-aware pools, the latter used only when Stage 1 has resolved a genuine category. Without that guard the default placeholder produces copy of the form “Your consumer product, Elevated”, which is how I found the problem: the fallback path had never run during development, and it only showed itself once the sweep put every product through it.

Stage 4 plans layout by rule, not by learning. The product occupies approximately 55 percent of the shorter canvas edge; portrait canvases centre it with copy above, and landscape canvases place it to the right with copy to the left along a Z-shaped reading path. I chose rules for reproducibility and ablatability, not because they produce the best layouts, and a learned alternative is an obvious extension. Testing them across extreme aspect ratios surfaced a limitation worth recording. Product boxes are integer pixels, so on extreme shapes the aspect ratio can be perturbed by up to one percent, and the fidelity measures cannot see that distortion at all, because Stage 6 records the already-distorted pixels as its reference.

3.5 Reproducibility infrastructure

Backgrounds and extractions are cached on disk under a digest of every input that determines them. The relighting level is deliberately left out of the background cache key, since Stage 6c never touches the backdrop; that one omission takes a full experimental grid from roughly 7.5 hours to 1.5. Runs resume at row granularity. I did not think that was worth building until a sweep was interrupted by system suspension twelve hours in, and then finished, on resumption, in four minutes.

Chapter IV

Method

4.1 Dataset

Twenty-four products were collected and stratified by the difficulty of separating them from their original background (Table 4.1).

TABLE 4.1: *Dataset strata by extraction difficulty.*

Stratum	n	Character
white	8	product photographed on a white sweep
plain	6	single-tone or gradient background
scene	6	styled shot with props and surfaces
transparent	4	pre-cut inputs; extraction error zero by construction

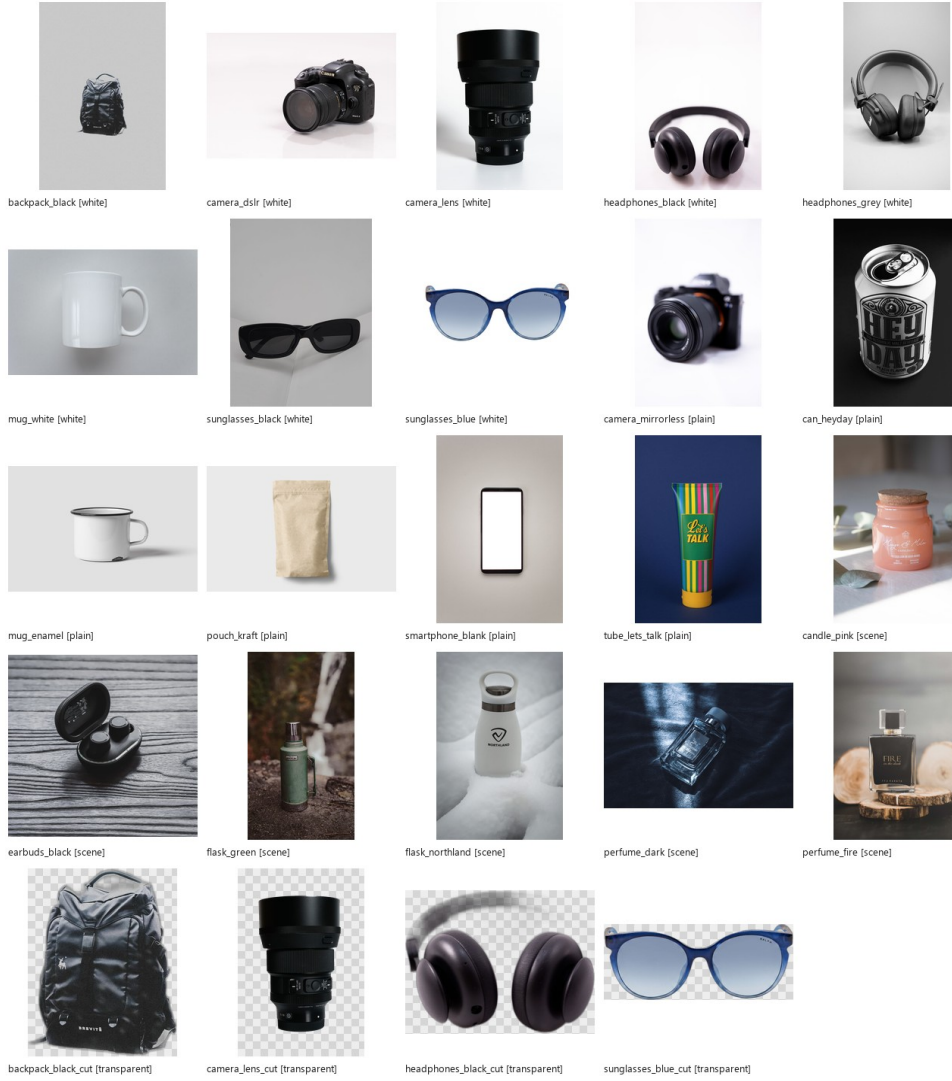


FIGURE 4.1: The 24-product dataset by stratum: white sweeps, plain backgrounds, styled scenes and the four pre-cut transparent items. Each transparent item shares its photograph with a white-stratum product, which is why the effective sample is twenty photographs (Section 6.2).



FIGURE 4.2: Stage 2 extraction examples. On the white stratum the cut-out is clean apart from the semi-transparent headband; on the scene stratum residual background survives around the bottle, which is the behaviour the stratification is designed to expose.

Each entry records its source URL and licence. I stratified on extraction difficulty because I expected it to dominate the fidelity results; pooling a clean pack shot with a styled scene would have hidden the effect. Products vary in material, with matte and specular surfaces roughly balanced, and most carry legible label text so the fine-detail measures have something to work on.

The transparent stratum required a methodological decision worth recording. Its purpose is to provide cases where Stage 2 introduces no error, so that any fidelity loss can be attributed to composition. Producing a mask independently of Stage 2 is the conventional route and I tried it first, building each one from a background distance transform, an Otsu threshold and a morphological clean-up, then checking the boundary at 1:1. Of four attempts one produced a usable mask, and the three failures were more informative than the success. A pair of sunglasses with lenses graduating to near-transparency could not be separated from a white background by any threshold, because the lower half of each lens genuinely resembles the backdrop. A pair of headphones absorbed its own contact shadow into the mask, the shadow being darker than the surface it fell on. A camera produced staircase artefacts from morphological closing at a kernel size appropriate for larger silhouettes.

The surviving mask is kept: `camera_lens_cut` is that item, and it is the only member of the stratum whose alpha was estimated rather than taken from the pipeline. For the remaining three I stopped estimating and constructed the case instead. The Stage 2 cut-out of a product already scoring 1.0 on the extraction screen is saved as an alpha channel and fed back in, so Stage 2 reads back exactly the mask it would have produced and contributes no error of its own. That is exact where a threshold is judged, and it sidesteps all three failure modes above.

Two things follow, and neither is incidental. These items isolate composition loss and carry no information about matting accuracy, because the mask is correct by definition rather than by comparison with the object. And every item in the stratum is the same photograph as a product in the white stratum, so each is sampled jointly with its source; Section 6.2 gives what that costs the effective sample.

4.2 Measures

Alongside SSIM (Wang *et al.*, 2004), PSNR, histogram correlation and LPIPS (R. Zhang *et al.*, 2018), four measures are used, two of them developed for this work.

Edge retention asks whether high-frequency detail survives a treatment. Gradient magnitudes are computed for the reference and the render within the product mask, the reference's strongest edges are selected by thresholding at the ninetieth percentile of its own gradient distribution, and retention is the proportion of those positions where the render preserves at least half the original magnitude. Thresholding against the reference instead of an absolute value keeps the measure scale-free: it asks whether detail that was present is still present, not whether detail exists.

Lighting coherence compares the direction in which brightness increases across the product with the same quantity for a ring of background surrounding it, mapping cosine similarity onto the unit interval. The direction is obtained by least-squares plane fit over interior pixels; Section 5.5 explains why the plane fit specifically, and the choice is not incidental. The background ring is sampled from an annulus around the product rather than the whole canvas, so that copy and other furniture do not contaminate the estimate.

Contact grounding compares mean luminance in a band beneath the product with the median of the remaining backdrop. It was introduced partway through the study to address a gap created by correcting lighting coherence. Its rationale is that an object resting on a surface occludes ambient light and casts a contact shadow, so the surface beneath it is darker than the same surface further away, whereas a cut-out simply pasted onto a backdrop exhibits no such gradient. I framed it as a property of photographs; framing it as a check that a shadow had been drawn would have made the ablation circular.

Extraction quality combines frame coverage, border contact and boundary colour contrast into a screen for failed segmentation. I included bounding-box fill at first and removed it after it flagged four correctly extracted products. A phone, a pouch and a can all legitimately fill their bounding boxes.

The two bespoke measures were validated internally before use, in the only way available without human raters: by checking that each responds when the component it is meant to detect is manipulated in a controlled way, and stays flat when components it should ignore are manipulated instead. That protocol establishes consistency with a measure's own definition and nothing stronger; Section 6.2 is explicit about the gap between the two. But it is also the protocol that caught both of the failures reported in Section 5.5, which is an argument for running it even when it cannot deliver external validity.

4.3 Experimental grid

The grid comprises 24 products, three seeds, two backends, seven relighting treatments and three output formats, together with a six-arm ablation, producing 5,184 rows with one row per rendered format.

4.4 Analysis

All contrasts are computed pairwise within (product, seed, backend, format). That is not just convention here. Lighting coherence varies far more between products than any treatment moves it, so an unpaired comparison would drown the effects of interest in between-product variance.

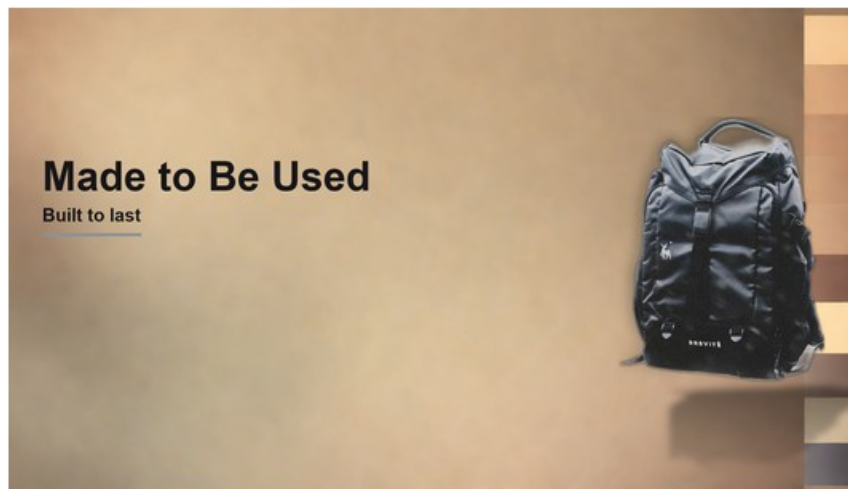
Effect sizes are reported alongside p-values throughout. With several hundred paired observations almost any systematic difference reaches conventional significance, so Cohen's d_z and the proportion of pairs improving are what indicate whether an effect is practically meaningful. Two of the results below are highly significant and negligible in size, and the distinction matters for how they should be read.

Chapter V

Results

5.1 Qualitative overview

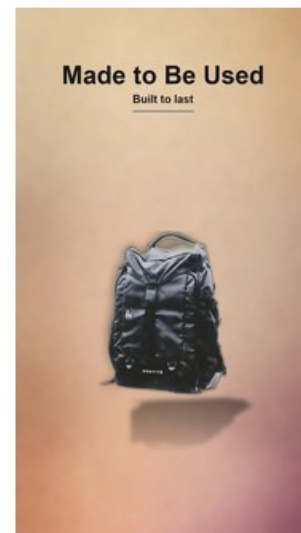
Before the quantitative results, this section shows what the pipeline actually produces, using renders drawn from the sample grid the experiment runner emits alongside the results table. All examples use the diffusers backend with relighting disabled, and the per-render measurements quoted beside them are taken from each sample's evaluation report. The copy is identical across every render for a reason worth stating: the headline resolves once per campaign through the provider chain of Section 3.4 and was held fixed throughout the experiments, so no variation reported anywhere in this chapter is attributable to the text.



banner 1920x1080



square 1080x1080



story 1080x1920

FIGURE 5.1: One campaign across the three output formats (*backpack_black*, *diffusers backend*, *relighting disabled*). The landscape banner places the product right with copy left; the portrait formats centre the product with copy above; all three canvases draw on the palette derived from the product in Stage 3.

Figure 5.1 carries a single product across the three formats. The layout rules of Section 3.4 are visible operating: the Z-path placement on the banner, and the centred stack on the portrait canvases. The campaign reads as a set rather than as three unrelated images because every canvas shares the brand palette extracted from the product. The generated backgrounds are not artefact-free: the banner render carries banded colour patches at its right edge, a failure of the background model rather than of the compositing, and it is in renders of this kind that the perspective tension discussed in Section 6.1 is easiest to see. A smaller detail also repays attention: the underline

beneath the tagline takes its colour from the same palette, which is the brand-consistency component of Section 3.3 doing its work at the scale of a few dozen pixels.



FIGURE 5.2: Single renders from three extraction strata: *backpack_black (white)*, *can_heyday (plain)* and *perfume_fire (scene)*; *diffusers backend*, *relighting disabled*.

Figure 5.2 shows one render from each of three strata, and what the stratification of Section 4.1 means in practice is visible at the silhouette. The white-sweep backpack and the plain-background can carry clean boundaries. Around the perfume bottle, traces of the styled source photograph survive extraction as a faint ghost of the original surroundings: the props and surfaces of the scene it was shot in, carried into the new backdrop as translucent residue. That residue is precisely the behaviour the stratification was designed to expose, and it is what the extraction-quality minima of Section 5.8 record numerically: the number and the artefact are the same phenomenon seen from two sides.

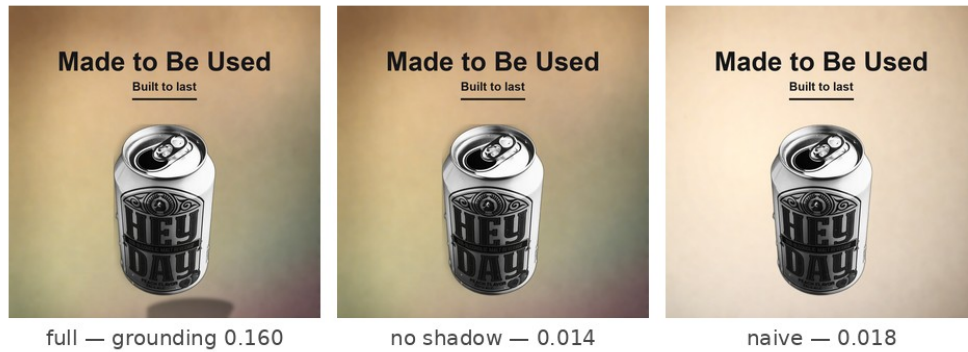


FIGURE 5.3: The same stimulus (*can_heyday*, square format, *diffusers* backend) under three ablation arms: the full pipeline, the contact shadow removed, and the naive paste-up. Contact grounding for this stimulus: 0.160 (full), 0.014 (no shadow), 0.018 (naive). Single-stimulus values, shown to illustrate the effect; the paired means across the full grid are in Table 5.2.

Figure 5.3 places one stimulus under three ablation arms. With the full pipeline the can rests on a visible darkening of the backdrop; with the shadow removed it floats; the naive arm loses the backdrop treatment as well and pastes the cut-out onto an untextured field. The measurements behave the way the paired analysis of Section 5.3 predicts, and more starkly, because a single comparison is not averaged against the between-product variance of the grid. Contact grounding falls from 0.160 under the full pipeline to 0.014 with the shadow removed — on this stimulus, removing that one component costs approximately the entire gap to the naive baseline at 0.018. Fidelity meanwhile cannot see the difference: SSIM reads 0.9941, 0.9941 and 0.9928 across the three arms, because everything that changed lies outside the product mask. The pair of observations is Section 5.3 and Section 5.5 in miniature. The shadow does most of the visible work, and the fidelity suite is structurally indifferent to it.

5.2 The fidelity and integration trade

Relighting strength was varied across five levels with the full pipeline and both backends (Table 5.1, Figure 5.4).

TABLE 5.1: Fidelity and scene-integration measures across relighting strengths.

Relight strength	SSIM	LPIPS	Edge retention	Lighting coherence
0.00	0.9884	0.0235	0.9467	0.5822
0.25	0.9879	0.0239	0.9476	0.5846
0.50	0.9872	0.0247	0.9483	0.5863
0.75	0.9861	0.0258	0.9486	0.5884
1.00	0.9845	0.0275	0.9487	0.5907

The relationship is monotonic in both directions, as the design predicted: as more relighting is applied, fidelity to the original product declines and agreement with the surrounding scene improves.

The magnitudes are asymmetric in a way the design did not predict. Comparing no relighting against full relighting pairwise, scene integration improves in roughly seven of every ten comparisons, but the improvement is small next to the variation between products ($dz = 0.23$). The fidelity cost is smaller in absolute terms and far more consistent, appearing in almost every comparison ($dz = -1.59$). In other words, you can rely on paying the cost and not on collecting the benefit.

This is weaker than an early single-product pilot had led me to expect. The pilot showed a coherence improvement of roughly 26 percent with an apparent optimum at intermediate strength, and neither survived at full dataset size. Section 6.1 examines the discrepancy. The smaller effect is the finding.

Broken down by stratum, the improvement concentrates where there is lighting structure for the product to agree with. These figures use the diffusers backend only: a procedural backdrop's brightness falloff is drawn from the same key light that drives the relighting, so coherence gains measured against it are partly self-agreement, and the generated backgrounds are the stricter test. Products photographed in styled scenes gain most in absolute terms ($\Delta = +0.0142$, $dz = 0.26$, $n = 54$), roughly four times the gain for products on white sweeps ($\Delta = +0.0038$, $dz = 0.33$, $n = 72$), and products on plain gradient backgrounds show no change at all ($\Delta = -0.0005$, $dz = -0.01$, $p = 0.93$). The two effect sizes are worth reading against each other: the scene stratum moves further but also varies more, so its dz is the smaller of the two despite the larger gain, and on this dataset neither is resolved well enough to rank the strata with any confidence.

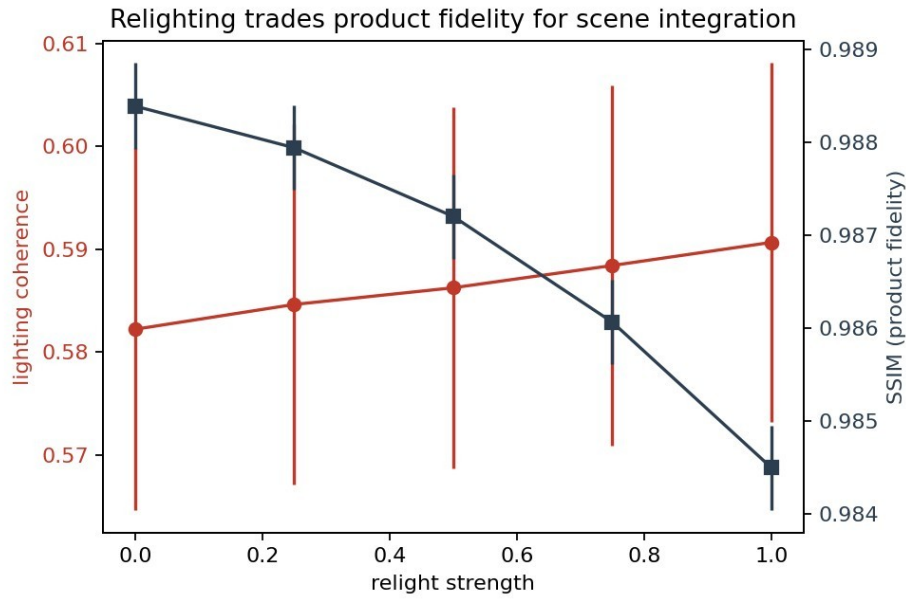


FIGURE 5.4: *Relighting strength against product fidelity (SSIM) and scene integration (lighting coherence). Error bars are standard errors over 432 paired observations.*

5.3 Component ablation

Five components were removed individually and in combination, and the resulting composites compared with the full pipeline using contact grounding (Table 5.2, Figure 5.5).

TABLE 5.2: *Ablation of compositing components, measured by contact grounding.*

Component removed	Change in grounding	Effect size (dz)	Pairs not improved
Contact shadow	−0.0708	−1.47	100%
Brand palette	−0.0639	−0.43	64%
Light coordination	−0.0039	−0.05	55%
Rim light	0.0000	—	—
All (naive baseline)	−0.1404	−0.89	96%

The contact shadow is the largest single contributor, and its effect is consistent: not one of the 432 paired comparisons ran the other way (411 worse, 21 exactly tied, none improved). It is several times the size of the relighting effect the study was designed to investigate, and shadow rendering is by far the less novel of the two techniques.

Removing the brand palette also reduced grounding, which I had not anticipated. The likely mechanism is that a neutral grey backdrop offers less tonal range for the darkening beneath the product to register against, so grounding depends partly on the backdrop carrying colour structure and

not only on a shadow being drawn. I have not established that. Testing it directly would mean holding the palette fixed while varying backdrop contrast, which I did not run.

The rim light shows no change on any measure. Section 5.5 explains why that is a limit of the measurement and not evidence that the rim light does nothing.

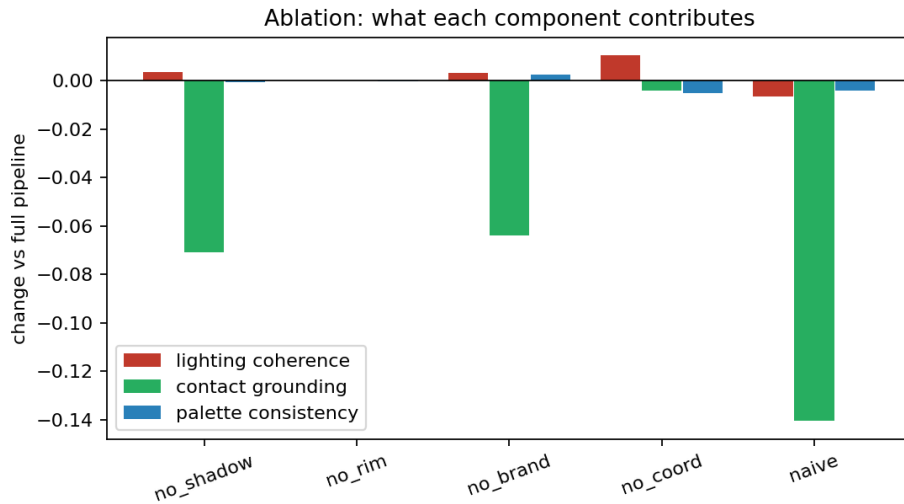


FIGURE 5.5: Change in each scene-integration measure when a compositing component is removed. Contact grounding separates the arms; lighting coherence cannot see components drawn outside the product silhouette.

5.4 Generative relighting

Diffusion image-to-image relighting was applied to the product region at a strength of 0.2 across the dataset. Table 5.3 and the paired statistics below are computed on the procedural backend, 216 pairs; the diffusers half shows the same pattern in every pair. Figure 5.6 makes the scale of the gap visible.

TABLE 5.3: Fidelity of the untreated, resample-control and generative arms.

Treatment	SSIM	LPIPS	Edge retention	PSNR
None	0.9934	0.0142	0.9627	38.1 dB
Resample control	0.9934	0.0142	0.9627	38.1 dB
Generative, strength 0.2	0.8217	0.1514	0.5224	23.4 dB

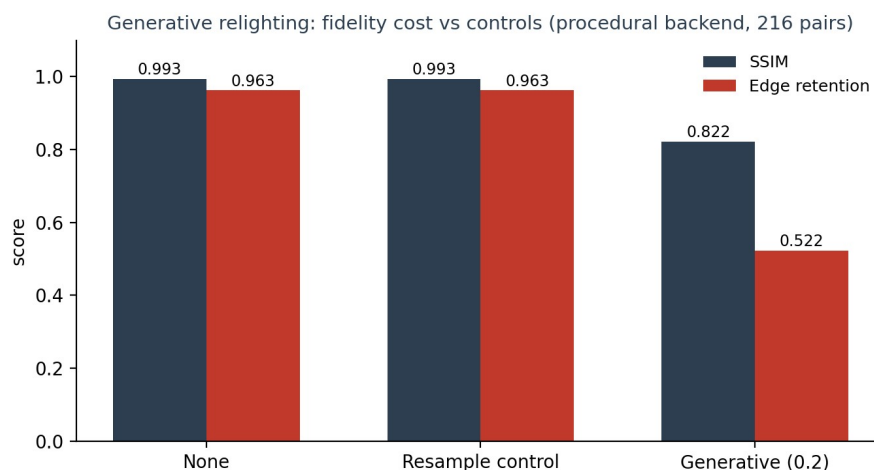


FIGURE 5.6: *The generative arm against the untreated and resample-control arms. The control is indistinguishable from no treatment, so the entire loss belongs to the model rather than to resampling.*

The generative method must operate below full resolution to fit within the available accelerator memory, which introduces a potential confound: some observed detail loss might be attributable to resampling rather than to the model. The resample arm was added to separate these, performing the identical resolution round trip with no model involved. Its results are identical to the untreated arm, because at the sizes the product occupies on the canvas the resolution bound is never actually reached. The detail loss therefore belongs to the model.

Roughly 44 percentage points of edge retention are lost, in all 216 paired comparisons without exception. Looking at the images shows why the effect is so uniform: the model does not confine itself to illumination, it reconstructs the region, and in one case I examined it produced legible label text that was not in the source image at all. For advertising that is a concern outside image quality. Markings that appear on a product and are not on the product are a legal problem, not an aesthetic one.

I explored lower strengths informally on a single product, where identity appeared to be lost even at 0.05, but I ran no systematic sweep below 0.2. What the data supports is that I found no acceptable operating point inside the range tested, which is not the same as saying none exists.

5.5 Three measurement problems and their resolution

The results above depend on measures I revised during the study. The revisions changed conclusions, so this section reports the process and not only the outcome.

A regional gradient average measures the boundary, not the region. My first implementation of lighting coherence estimated the product's shading direction by averaging the numerical gradient over the product region. That seemed like the obvious thing to do, and it is wrong. By the divergence theorem the integral of a gradient over an area is fixed by its boundary, so what the average actually reported was the luminance step between the product and whatever lay immediately outside it, with the shading inside the product contributing almost nothing.

The consequence was large. Painting a rim light along the silhouette, which is drawn entirely outside the product, shifted the measured product lighting direction by roughly 0.10, about four times the size of the entire relighting effect under study. Run with this measure, the ablation reported that removing the rim light improved coherence and that the naive baseline beat the full pipeline by a wide margin. Both conclusions were artefacts of the estimator.

The ablation is what caught it. A result that contradicts the design that strongly is usually the measure's fault, and it was. Replacing the regional average with a least-squares plane fit over eroded interior pixels dropped rim-light sensitivity to zero, and I repeated the affected experiments. The superseded data is kept as `experiments/results_broken_metric.csv`.

The correction created a blind spot. The corrected measure reads interior pixels only. Both the contact shadow and the rim light are drawn outside the silhouette, so the fix made them invisible to it by construction. The ablation went quiet: five arms returning values indistinguishable from zero.

This is worth stating carefully, because a table of null results reads naturally as evidence that the components do not matter, and that is not what it showed. Nothing in the measurement suite was looking at them. I added contact grounding to close part of the gap, and with it the shadow's contribution became measurable and large, as Section 5.3 reports.

Contact grounding then needed a correction of its own. Its first version compared a narrow annulus hugging the silhouette against a wider one, and reported that adding a shadow reduced grounding. The projected shadow reaches considerably further than the narrow annulus, so it darkened the comparison region while leaving the sample region alone, and the rim light brightened the sample on top of that. Both bands have to sit where the effect physically falls, so I measured where it falls instead of assuming.

Fidelity measures cannot detect extraction failure. On one styled-scene product, GrabCut returned a region containing the backdrop and the props

instead of the product. Every fidelity measure stayed near its ceiling: SSIM above 0.998, palette consistency above 0.95.

The reason is structural, not specific to my implementation. SSIM compares the rendered product region against the Stage 6 reference, and that reference is itself derived from the extraction. When the extraction fails, the reference records the failure, and the comparison then confirms that the failure was faithfully preserved. Any evaluation that references an intermediate pipeline artefact instead of an independent ground truth is exposed to this, and the numbers give no warning.

I wrote the extraction quality screen in response. Across the twenty products it applies to (the transparent stratum is excluded, for the reason Section 5.8 gives), it separates the two matting methods clearly: 0.916 for the U²-Net-based method against 0.746 for GrabCut on the same twenty. It does not, however, catch the original mid-frame failure that prompted it, because that failure neither reached the frame edges nor claimed an unusual share of the image. So it is documented as a filter for choosing cases to inspect, not as an automatic gate, and the limitation is recorded in `metrics.py` as well as here.

I also tried to substitute machine discrimination for human rating, training a classifier to separate each ablation arm's composites from photographs on the reasoning that a more realistic composite should be harder to detect. It did not work. Accuracy ran from 0.83 to 0.92 across all six arms with substantially overlapping confidence intervals, and the full pipeline was no harder to detect (0.854) than the naive baseline (0.875). Two explanations are likely. Forty-eight samples per arm cannot resolve differences of two or three percentage points; and the signal available to the classifier looks dominated by pipeline provenance, which every arm shares, and not by compositing quality, which is the thing that differs. I report the attempt because it bounds what can stand in for human evaluation.

The three problems share a shape worth naming. None of them announced itself as an error; each produced numbers that were plausible individually and only failed against something else: the ablation contradicting the design, a null table contradicting a visible component, a ceiling-level score contradicting a render that was plainly wrong. What caught them, in every case, was redundancy: a second reading of the same system that the broken measure had to agree with and did not. The practical lesson is that an evaluation framework for generative pipelines needs to be built with its own contradictions on purpose, with arms whose expected ordering is known in advance and stimuli whose correct score is fixed by construction, because a measure that is never given the opportunity to disagree with anything can be wrong indefinitely.

5.6 An interaction between distillation and negative prompting

SDXL-Turbo operates at guidance scale zero, where classifier-free guidance performs no interpolation between conditional and unconditional predictions and the unconditional branch is never evaluated. A negative prompt supplied on this path has no effect.

This follows from the distillation procedure (Sauer *et al.*, 2023); it is not a misconfiguration, and it appears not to be prominently documented anywhere I looked. It matters here in particular, because the negative prompt is the usual way to keep a competing product out of the canvas region reserved for the real one, and nothing in the generated images reveals that it has stopped working.

The behaviour was confirmed by running full SDXL at guidance 6.5, where photographic equipment reliably disappeared from backgrounds that had otherwise contained it, at approximately eleven times the generation cost.

Two smaller observations about prompt construction came out of the same diagnosis. Negations in the positive prompt were counter-productive: “no floor line” supplies the tokens “floor” and “line”, and the feature became more likely, not less. Architectural vocabulary summoned its context, with the word “wall” reliably producing the skirting board, door and floor of a room. Both behaviours are consistent with how the CLIP text encoders (Radford *et al.*, 2021) that condition SDXL represent such prompts, and both reproduced across all three seeds.

5.7 Backend comparison

TABLE 5.4: *Fidelity and brand consistency by background backend.*

Backend	SSIM	LPIPS	Edge retention	Palette consistency
Procedural	0.9934	0.0142	0.9627	0.9266
Diffusion	0.9834	0.0328	0.9307	0.8825

The deterministic procedural backend performs better on every fidelity measure and on brand consistency. The brand result is largely definitional, since the procedural background is built from the brand palette. The fidelity difference is not, and Section 6.1 takes it up.

5.8 Extraction quality across strata

TABLE 5.5: *Extraction quality by stratum.*

Stratum	n	Mean quality	Minimum
plain	6	0.950	0.699
white	8	0.933	0.468
scene	6	0.857	0.145

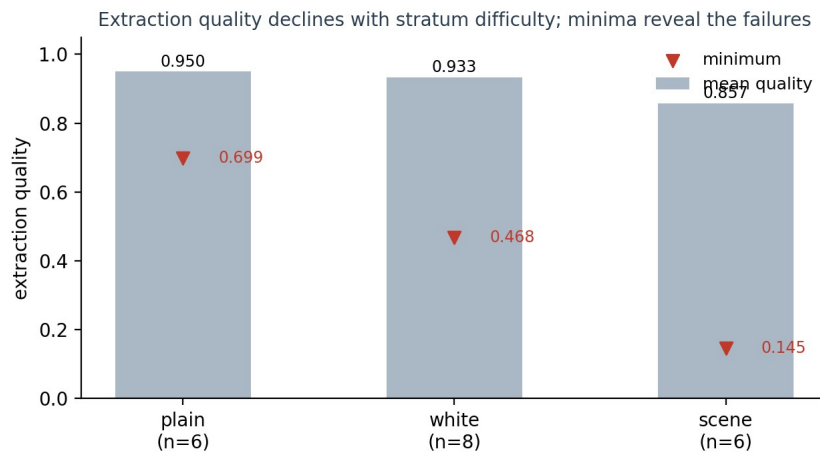


FIGURE 5.7: *Mean extraction quality by stratum with per-stratum minima. The minima carry the information: both sub-threshold products sit far below their stratum means.*

Quality declines in the direction the stratification anticipated. The minima say more than the means. Two products fell below the review threshold, and inspection found one genuine residual artefact and one false positive: a white mug photographed on a white sweep, where boundary contrast is genuinely absent and the screen cannot tell a missing signal from a poor mask.

The transparent stratum is excluded from this table because the screen does not apply to it. A pre-cut input has no background pixels outside the mask, so boundary contrast is undefined and computes to approximately zero, dragging the composite score down even though the mask is exact by construction. Printed without that caveat, the number would tell a reader the opposite of what is true.

Chapter VI

Discussion

6.1 Interpreting the results

Why the relighting effect is small. The pilot suggested a substantial improvement and the full dataset shows a modest one. Two things appear to explain the gap. The pilot used a single product photographed in a styled scene, which is the stratum where the effect is largest; had I picked a product on a plain gradient background instead, the same pilot would have told me just as confidently that the treatment does nothing. Second, the pilot ran before I corrected lighting coherence, and the corrected measure credits the treatment with less. There is also a third, more general reading: a pilot is a single-stimulus estimate, and single-stimulus estimates in this study ran large wherever they were examined; the Figure 5.3 grounding gap is several times the paired mean for the same contrast. Effects measured before averaging over between-product variance should be expected to shrink at scale, and the pilot's 26 percent was not an anomaly so much as a preview of that shrinkage.

The practical implication is that parametric relighting is worth applying where the background has genuine lighting structure and not worth its fidelity cost where it does not. The pipeline already knows which method produced the backdrop and could estimate whether that backdrop carries directional structure, so this is a policy someone could implement, not just an observation.

Why the shadow matters more than the relighting. This surprised me, since the relighting was the designed contribution and the shadow a supporting component. It fits how the two operate, though. The relighting adjusts brightness across the product's interior, a change spread over many pixels and small at each one. The shadow introduces a structure that was otherwise entirely absent, since a pasted cut-out has nothing beneath it, and putting a plausible contact shadow there shifts the local image statistics a long way. It also fits the compositing literature's long-standing emphasis on shadow and contact cues, and the observation in harmonisation work (Cong *et al.*, 2020) that composite realism depends heavily on the junction between foreground and background, not on the foreground alone.

What the qualitative renders add. Figure 5.3 gives the mechanism a visual form. A cut-out without a shadow does not read as an object in the scene at all; it reads as a sticker on top of one, and no interior adjustment of the product can repair a relationship that is missing beneath it. This is also why

the naive baseline's deficit is nearly exhausted by the shadow alone on that stimulus: most of what the full pipeline adds that a paste-up lacks is concentrated at the contact region, which is where a viewer's evidence about weight and placement lives. The relighting, by contrast, spends its fidelity budget spreading a small change across the product's interior, and the asymmetry between the two is the asymmetry Section 5.3 measures.

Why the procedural backend performs better on fidelity. Not because the product is treated differently; it is handled identically in both cases. The fidelity measures are computed over the product region of the composite, and a busier, higher-frequency background produces more boundary interaction at the silhouette. Something more interesting turned up while I was looking through the images. A generated background that carries its own perspective puts the product's implied camera at odds with the scene's, and no amount of shadow or colour adjustment hides it. A gradient sweep has no perspective to contradict, which may be part of why commercial product photography leans on one so heavily, and it suggests that for this application a generative background is worth more for its texture and colour than for any scene it depicts.

Why generative relighting behaves as it does. Diffusion image-to-image reconstructs the region it operates on from a partially noised latent. It has no mechanism for separating the illumination of a surface from the markings on it, so asking it to change the lighting is in effect asking it to regenerate, and identity-bearing detail such as small text is exactly the kind of high-frequency content a generative prior is least likely to reproduce. Methods built specifically for relighting, such as IC-Light (L. Zhang *et al.*, 2025), impose stronger structural conditioning and should fare better. Nothing here is a claim about relighting in general, only about the image-to-image approach I tested, on this dataset, at the strengths I examined.

6.2 What the results do not support

Scale. Twenty-four products from a single stock library, with matte branded packaging underrepresented, is a small and non-representative sample. The stratified breakdowns rest on between 36 and 72 paired observations each: adequate for the large effects reported, marginal for the small ones. The relighting result in particular sits at the edge of what this dataset can resolve, and should be read as indicative. All four items in the transparent stratum are the same photographs as products in the white stratum, carrying an alpha channel instead of a flattened background, and each is sampled jointly with its source. The dataset therefore rests on twenty distinct photographs rather than twenty-four, which reduces the effective sample further than the headline count suggests.

Absence of human evaluation. No human raters were involved at any point. Every statement about how an image looks is therefore mediated by automated measures, two of which I wrote for this study. I validated those two internally, checking that they respond to controlled manipulation of the components they are meant to detect and stay flat under manipulations they should ignore. That establishes only that a measure behaves consistently with its own definition. It cannot establish that the quantity it captures is one a viewer would care about.

The limitation is not incidental, given that two measures in this project turned out to be capturing something other than what I intended. I cannot rule out from inside the framework that a third is doing the same. Claims are accordingly phrased in terms of what the measures resolve, not what a person would notice, and where I invoke LPIPS its validity is borrowed from the human data it was calibrated on. A two-alternative forced-choice study addressing this is designed and implemented (five comparisons, about twelve minutes a rater, including the one that would settle the rim light question), but I did not run it within the scope of this project.

Unmeasured components. The rim light remains unmeasured. Contact grounding covers the shadow; edge treatment needs either a boundary-region measure or human rating, and the machine-discrimination substitute described in Section 5.5 was not adequate. These results therefore support no conclusion at all about the value of edge treatment in this pipeline.

Range of the generative comparison. The generative relighting arm was run at a single strength across the full dataset. The finding is that I found no acceptable operating point within the range examined. That is narrower than a claim that none exists, and narrower still than any claim about relighting methods other than the one I tested.

Qualitative examples. The renders in Section 5.1 were selected to be legible, not to be representative, and the per-stimulus numbers quoted there are larger than the paired means for the reason stated in the captions: a single comparison is not averaged against the between-product variance that dominates the grid. Nothing in Section 5.1 is evidence beyond illustration. Every quantitative claim in this dissertation rests on the paired analysis of Sections 5.2 to 5.8, not on the examples.

Statistical assumptions. The paired design controls the dominant source of noise, but each product contributes multiple rows through seeds and formats, and those rows are not independent. The reported p-values consequently overstate confidence, by an amount I have not quantified. Effect sizes and proportions of improved pairs do not rest on independence assumptions, and they are the figures to trust throughout.

Negative results retained. Four of the results above are negative: the pilot effect that did not replicate at scale, the ablation that was uninformative because the measure was blind, the machine-discrimination substitute that could not separate the arms, and three of the four hand-masking attempts. Each one bounds what the framework can conclude, which is why they are here. The data produced under the uncorrected measure is kept alongside the corrected data so the artefact can be inspected and not just described.

6.3 Relation to prior work

The measurement problem identified in Section 5.5 has a general form that may be relevant beyond this system. Any pipeline evaluating a late stage against a reference produced by an earlier stage is exposed to the same failure, and pipelines of this shape are common in generative image editing, where an extraction, a mask or a depth estimate frequently serves as the reference for everything downstream. The harmonisation literature largely avoids this by construction, since benchmarks such as iHarmony4 (Cong *et al.*, 2020) pair composites with the real images they were derived from and therefore have access to genuine ground truth. Where such pairing is unavailable, as it is for real product photographs which have no correctly-composited counterpart, the exposure returns.

The trade-off quantified here also sits in a specific relation to harmonisation. That literature seeks to modify the foreground until it fits its new background, and measures success by how well it fits. This work begins from a constraint forbidding exactly that modification, and the relighting stage is a bounded relaxation of it. My results suggest that for advertising the useful region of this space lies much closer to the preservation end than harmonisation methods typically operate, and that most of the achievable integration comes from what you put around the product rather than from what you do to it. That is a suggestion drawn from one dataset and one pipeline, not a general claim about either literature.

Read against Table 2.1, the trajectory of the harmonisation literature is a steady loosening of what may be done to the foreground: matched statistics first, then learned appearance, and most recently generative regeneration of the composited region. The generative relighting result of Section 5.4 is what the far end of that trajectory looks like when the foreground carries legal identity: the same capacity that lets a model harmonise convincingly is the capacity that lets it fabricate label text. The constraint this dissertation starts from is therefore not a conservative preference but a boundary that the rest of the trajectory cannot cross for this application, and the useful question becomes how much integration is achievable while standing behind it.

Chapter VII

Conclusion

This dissertation set out to build a system that generates advertising backgrounds while preserving product pixels exactly, and to work out what that constraint costs. Building it was straightforward. Working out the cost was not: the preservation guarantee makes conventional fidelity measures uninformative by construction, and the measures I needed for the other half of the question turned out, twice, to be responding to something other than what I had named them for.

Within the dataset examined, three findings emerged. Diffusion image-to-image relighting altered product appearance to a degree that appears difficult to reconcile with advertising requirements, at the strength tested, and did so consistently enough that the result is unlikely to be an artefact of this particular sample. The contact shadow contributed more to composite grounding than any other component examined, including the relighting the study was designed around. Parametric relighting produced a small and consistent improvement in scene integration at a small and highly consistent fidelity cost, concentrated in cases where the background carries directional lighting structure.

For a practitioner the results reduce to three pieces of guidance. Draw the contact shadow from the product's own silhouette rather than an ellipse, because it is the largest single contributor measured here. Spend the relighting budget only where the backdrop carries directional lighting structure, because that is the only place the benefit was observed. And do not put identity-bearing pixels through an image-to-image model at any strength examined in this study, because the cost was uniform, large, and in one observed case legally consequential.

The methodological findings probably transfer further than the empirical ones. That a regional gradient average is determined by its boundary is a property of the estimator, not of this system. That correcting a measure can leave it blind to the very components you want to compare is a hazard of iterative metric development generally. That fidelity measures computed against a pipeline-internal reference cannot detect corruption of that reference is structural, and applies to a broad class of generative editing systems. I established all three by controlled manipulation, with no appeal to human judgement, which is why I can state them more confidently than any of the effect sizes.

The most significant limitation is the absence of human evaluation, which leaves every appearance-related claim resting on measures validated only against themselves. The study that would address it is specified and implemented but was not run in the project's timeframe, and it is the first thing I would do next. After that: a boundary-region measure would close the remaining gap in the ablation, a larger and better balanced dataset would tighten the small-effect estimates, and relighting methods with explicit identity conditioning are worth putting through the same framework, since what I found here concerns one approach and not the objective in general.

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